

VAR1 — Power Plant Steam Turbine Optimization

BT2024167 | Polynomial Regression Report

1. Problem Description

VAR1 predicts the plant's Net Power Score (y) from six turbine operating variables: $x1$ high-pressure steam valve adjustment, $x2$ condenser coolant flow rate adjustment, $x3$ re-injection pump hydraulic pressure, $x4$ turbine blade pitch angle, $x5$ non-condensable gas exhaust valve rate, and $x6$ steam inlet pressure adjustment. The task is to fit a polynomial regression model using the supplied personalized training data and predict the target for the supplied test inputs.

2. Dataset Description

The personalized VAR1 training set contains 1,000 rows and 7 columns (six inputs and target y); the test set contains 1,000 rows and the same six input columns. All columns are numeric. There are no missing values, no duplicate training rows, and 8 duplicate feature rows in the test set. Each input ranges from -1 to 1 in the training data. The training target ranges from -7.8732 to 11.8996 (mean 0.8529). No ID or index feature was present.

3. Methodology

An 80/20 train-validation split was created from the training data with **random_state=42**. For each degree from 1 through 10, the model used `PolynomialFeatures(degree=d, include_bias=False)` followed by `LinearRegression`. Validation Mean Squared Error (MSE) and R^2 were used to compare the models. The hidden test data was not used to select the degree or tune the model. No additional scaling was applied because the inputs were already bounded between -1 and 1 .

4. Polynomial Degree Selection

Table 1 reports the measured validation results. Degree 1 has substantially higher error, consistent with underfitting this data. Performance improves through degree 4, which gives the lowest validation MSE and highest validation R^2 . Degree 5 performs worse, degree 6 is highly unstable on this split, and degrees 7–10 do not recover degree 4's validation performance.

Degree	Polynomial Features	Validation MSE	Validation R^2
1	6	9.7544818	0.12564193
2	27	3.7625869	0.66273470
3	83	1.0576628	0.90519475
4	209	0.85152215	0.92367249
5	461	1.6821712	0.84921598
6	923	208.77378	-17.713761
7	1715	10.184963	0.08705512
8	3002	7.1964290	0.35493694
9	5004	4.6477478	0.58339193
10	8007	4.6166547	0.58617901

Table 1. Validation performance on the held-out 20% training-data split.

VAR1 — Model Validation and Submission

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5. Five-Fold Cross-Validation

A shuffled five-fold cross-validation check (`random_state=42`) was run for the three strongest candidates from the split validation: degrees 3, 4, and 5. Degree 4 also has the lowest mean CV MSE and highest mean CV R^2 among these candidates. This provides additional support for the degree-4 choice. The train MSE values are included for context; they are not used as hidden-test estimates.

Degree	CV MSE Mean	CV MSE Std	CV R^2 Mean	CV R^2 Std	Train MSE Mean
4	0.78515636	0.15831600	0.91834494	0.01464419	0.35081375
3	0.95905710	0.10305704	0.89993865	0.01022374	0.74457012
5	1.69058160	0.50668384	0.82305742	0.05574517	0.12208108

6. Final Model and Prediction File

Based on the validation and cross-validation evidence, degree 4 was selected. The final pipeline, `PolynomialFeatures(degree=4, include_bias=False)` plus `LinearRegression`, was refit using all 1,000 available training samples. It generated one prediction for each of the 1,000 test rows, preserving their original order. Predictions were saved as **BT2024167_pred_var1.csv**. The submission contains exactly one column, `y`, with no index column.

The selected model's held-out validation scores were $MSE = 0.85152215$ and $R^2 = 0.92367249$. These are validation results only; the test target is hidden, so no test MSE or test R^2 is reported.

7. Conclusion

Degree 4 was selected because it achieved the best validation MSE and R^2 , and the five-fold check also favored it over degrees 3 and 5. Higher-degree models did not improve validation performance and showed substantial instability at degree 6. The degree-4 polynomial regression model was therefore refit on all training data to produce the VAR1 submission predictions.